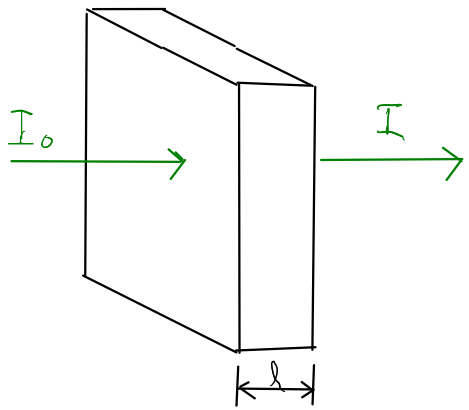
Tauc Plot

Beer Lambert's law :



$$\frac{I}{I_0} = e^{-\alpha l}$$

Absorption coefficient

$$[\alpha] = cm^{-1}$$

Absorbance :

$$A = \log\left(\frac{I_0}{I}\right)$$

$$A = -\log\left(\frac{I}{I_0}\right)$$

$$A = -\log T$$

Transmitance :

$$T = \frac{I}{I_0}$$

Beer Lambert's law :

$$\frac{I}{I_0} = e^{-\alpha l}$$

$$-\log\left(\frac{I_0}{I}\right) = -\alpha l \log e$$

$$A = \alpha l \log e$$

$$\alpha = \frac{A}{l \log e}$$

$$\boxed{\alpha = 2.303 \frac{A}{l}}$$

Tauc Plot method:

$(\alpha h\nu)^n = K(h\nu - E_g)$

$n = 2$, direct allowed transition

$n = 1/2$, indirect allowed transition

$n = 2/3$, forbidden direct transition

$n = 1/3$, forbidden indirect transition

$$\left[\left(\frac{A}{l \operatorname{Log} e}\right) h\nu\right]^n = K(h\nu - E_g)$$

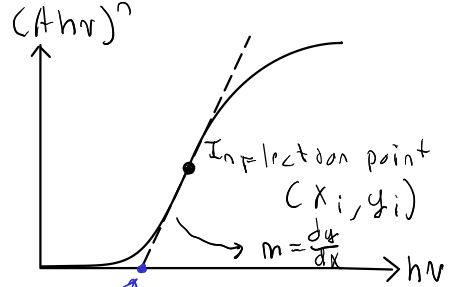
$$\frac{(A h\nu)^n}{(l \operatorname{Log} e)^n} = K(h\nu - E_g)$$

$$\frac{(A h\nu)^n}{K(l \operatorname{Log} e)^n} = h\nu - E_g$$

$$\frac{1}{K(l \operatorname{Log} e)^n}(A h\nu)^n = h\nu - E_g$$

When $(A h\nu)^n = \emptyset$

$h\nu = E_g$



$y_i = m x_i + b$

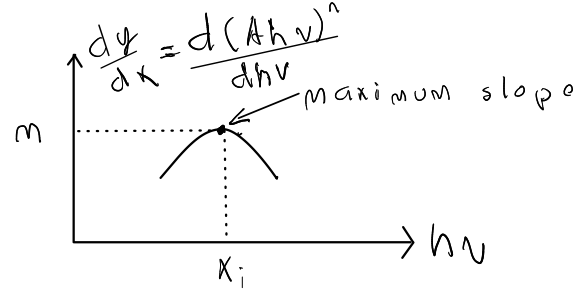
$b = y_i - m x_i$

When: $(A h\nu)^n = 0 \rightarrow h\nu = E_g$

$\emptyset = m E_g + b$

$\emptyset = m E_g + (y_i - m x_i)$

$$\boxed{E_g = x_i - \frac{y_i}{m}}$$

$\rightarrow y = m x + b$

$$\boxed{y = m x + (y_i - m x_i)}$$